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# Day 3 - NumPy Transaction Analysis

import numpy as np

# Create an array of 20 sample transaction amounts
transactions = np.array([
    250, 480, 1250, 750, 3200,
    890, 1500, 450, 2750, 1100,
    650, 4200, 980, 1850, 560,
    3100, 720, 2400, 1350, 5000
])

print("Transaction Amounts:")
print(transactions)

# Find the top 5 transaction values
top_5 = np.sort(transactions)[-5:][::-1]

print("\nTop 5 Transaction Amounts:")
print(top_5)

# Apply 18% tax using broadcasting
tax_rate = 0.18
transactions_with_tax = transactions * (1 + tax_rate)

print("\nTransactions After Adding 18% Tax:")
print(transactions_with_tax)

# Calculate basic statistics
mean_value = np.mean(transactions)
max_value = np.max(transactions)
min_value = np.min(transactions)

print("\nStatistics:")
print("Mean:", mean_value)
print("Maximum:", max_value)
print("Minimum:", min_value)
```

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Transaction Amounts:
[ 250  480 1250  750 3200  890 1500  450 2750 1100  650 4200  980 1850
 560 3100  720 2400 1350 5000]

Top 5 Transaction Amounts:
[5000 4200 3200 3100 2750]

Transactions After Adding 18% Tax:
[ 295.   566.4 1475.   885.  3776. 1050.2 1770.   531.  3245.  1298.
 767.  4956. 1156.4 2183.   660.8 3658.   849.6 2832.  1593. 5900. ]

Statistics:
Mean: 1671.5
Maximum: 5000
Minimum: 250
```

Day 3 — NumPy Explanation

NumPy is a Python library used for numerical and array-based computation.

What I Practiced

- Creating a NumPy array
- Indexing and sorting array values

- Mathematical operations on arrays
- Broadcasting
- Mean, maximum, and minimum

What I Built

I created a NumPy array containing 20 sample transaction amounts.

I then:

1. Found the top 5 transaction amounts.
2. Applied an 18% tax to all transactions using a vectorized NumPy operation.
3. Calculated the mean transaction amount.
4. Found the maximum transaction amount.
5. Found the minimum transaction amount.

Why NumPy is Useful

NumPy allows numerical operations to be performed efficiently on entire arrays instead of processing values one by one with manual Python loops.

✓ Day 3 — Conclusion

Today I learned how to create and manipulate NumPy arrays.

I used NumPy to perform calculations on transaction data efficiently. I also practiced sorting, mathematical operations, broadcasting, and basic statistical functions.

Learning Outcome

I can create NumPy arrays and use vectorized operations and broadcasting to process numerical data.

Start coding or [generate](#) with AI.